



# CPSC 100

# Computational Thinking

## Artificial Intelligence

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# How does NLP work?



# NLP in a nutshell

- NLP draws on many disciplines: linguistics, cognitive science, psychology, logic, computer science, philosophy, engineering.
- **Traditional approach:** Long list of rules for processing language, formulated by people and programmed into computers
- **Modern approach:** Machines learn from text examples using artificial neural networks and similar approaches. Statistical methods allows to compare different interpretations



# Traditional NLP Steps

1. **Recognize speech** (typically chatbots receive ASCII versions of the questions)
2. **Syntax analysis, or parsing**: inferring parts of speech and sentence structure, using a lexicon and grammar
3. **Semantic analysis**: inferring meaning using syntax and semantic rules
4. **Pragmatics**: inferring meaning from contextual information



# Limitations of traditional NLP

- Natural language is structurally **ambiguous**, so parsing alone cannot lead to understanding.
- Synonyms for words can't be used interchangeably in every context, e.g., “*minister of agriculture*” isn't “*priest of farming*.”
- Natural languages have many exceptions to grammatical rules; there's no agreed-upon grammar for all uses of a language.

# Parsing is the basis for programming

- A computer has to “**understand**” programs in order to execute them
- Programming languages are designed so that they can be parsed **unambiguously**
- A grammar specifies all the possible programs that can be written in a language
- **Designing programming** languages (and their grammars) is a fun and important part of computer science

# NLP: Semantic Analysis Processes

- **Word Sense Disambiguation:** Identifying the correct meaning of words with multiple interpretations in a given context.
- **Semantic Role Labeling:** Assigning roles (e.g., agent, patient) to elements in a sentence to understand their relationships.
- **Named Entity Recognition:** Identifying and categorizing named entities like names, locations, and organizations in text.
- **Semantic Parsing:** Converting natural language into a formal representation of meaning, aiding in understanding user queries or commands.
- **Sentiment Analysis:** Determining the sentiment expressed in text (positive, negative, or neutral).
- **Semantic Similarity:** Measuring the likeness or relatedness of text based on its meaning.

# Tradition vs. Modern

- **Parsing:** While traditional parsing involves breaking down and analyzing the structure of language (like sentence structure and grammar), ChatGPT's transformer architecture processes text through a series of layers that capture different aspects of language, including some structural elements.
- **Semantic Analysis:** ChatGPT performs semantic analysis through its deep learning model, which understands the meanings and relationships of words and phrases in context. This is essential for generating coherent and contextually relevant responses.



(1) Input: Sentence: “The rat ate cheese”

# ASCII TABLE

| Decimal | Hex | Char                   | Decimal | Hex | Char    | Decimal | Hex | Char | Decimal | Hex | Char  |
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| 0       | 0   | [NULL]                 | 32      | 20  | [SPACE] | 64      | 40  | @    | 96      | 60  | `     |
| 1       | 1   | [START OF HEADING]     | 33      | 21  | !       | 65      | 41  | A    | 97      | 61  | a     |
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| 5       | 5   | [ENQUIRY]              | 37      | 25  | %       | 69      | 45  | E    | 101     | 65  | e     |
| 6       | 6   | [ACKNOWLEDGE]          | 38      | 26  | &       | 70      | 46  | F    | 102     | 66  | f     |
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| 9       | 9   | [HORIZONTAL TAB]       | 41      | 29  | )       | 73      | 49  | I    | 105     | 69  | i     |
| 10      | A   | [LINE FEED]            | 42      | 2A  | *       | 74      | 4A  | J    | 106     | 6A  | j     |
| 11      | B   | [VERTICAL TAB]         | 43      | 2B  | +       | 75      | 4B  | K    | 107     | 6B  | k     |
| 12      | C   | [FORM FEED]            | 44      | 2C  | ,       | 76      | 4C  | L    | 108     | 6C  | l     |
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| 16      | 10  | [DATA LINK ESCAPE]     | 48      | 30  | 0       | 80      | 50  | P    | 112     | 70  | p     |
| 17      | 11  | [DEVICE CONTROL 1]     | 49      | 31  | 1       | 81      | 51  | Q    | 113     | 71  | q     |
| 18      | 12  | [DEVICE CONTROL 2]     | 50      | 32  | 2       | 82      | 52  | R    | 114     | 72  | r     |
| 19      | 13  | [DEVICE CONTROL 3]     | 51      | 33  | 3       | 83      | 53  | S    | 115     | 73  | s     |
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| 21      | 15  | [NEGATIVE ACKNOWLEDGE] | 53      | 35  | 5       | 85      | 55  | U    | 117     | 75  | u     |
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| 31      | 1F  | [UNIT SEPARATOR]       | 63      | 3F  | ?       | 95      | 5F  | _    | 127     | 7F  | [DEL] |

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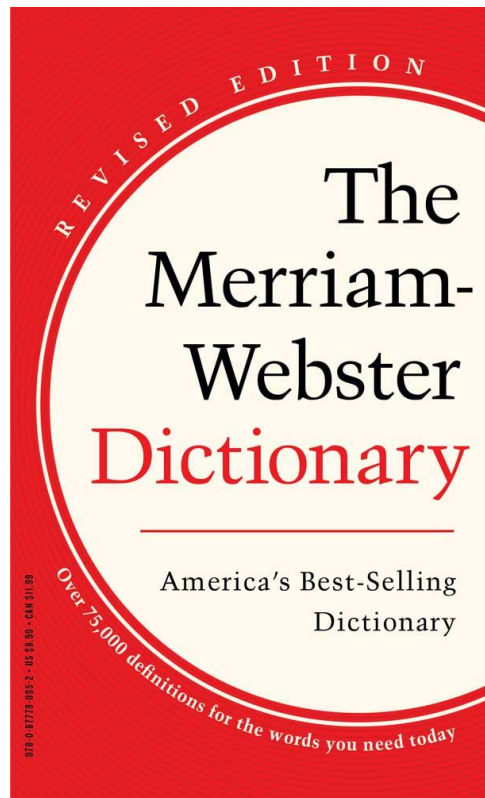
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(2) Lexicon:

| rat    | Noun    |
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| cheese | Noun    |
| ate    | Verb    |
| The    | Article |



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(3) Grammar:

Sentence → NounPhrase, VerbPhrase

“The dog barked.”

- Noun Phrase (NP): “The dog”
- Verb Phrase (VP): “barked”

“A cat is sleeping.”

- Noun Phrase (NP): “A cat”
- Verb Phrase (VP): “is sleeping”

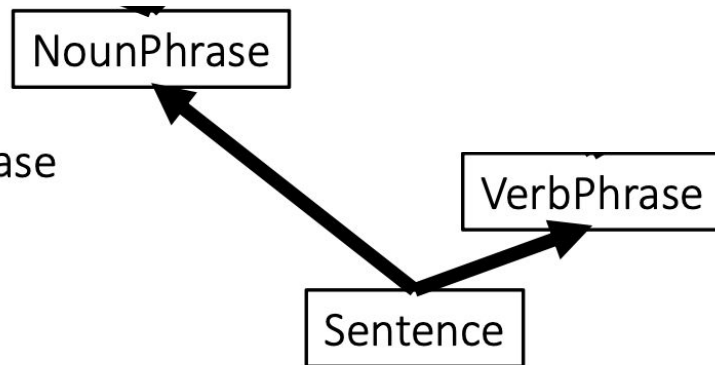
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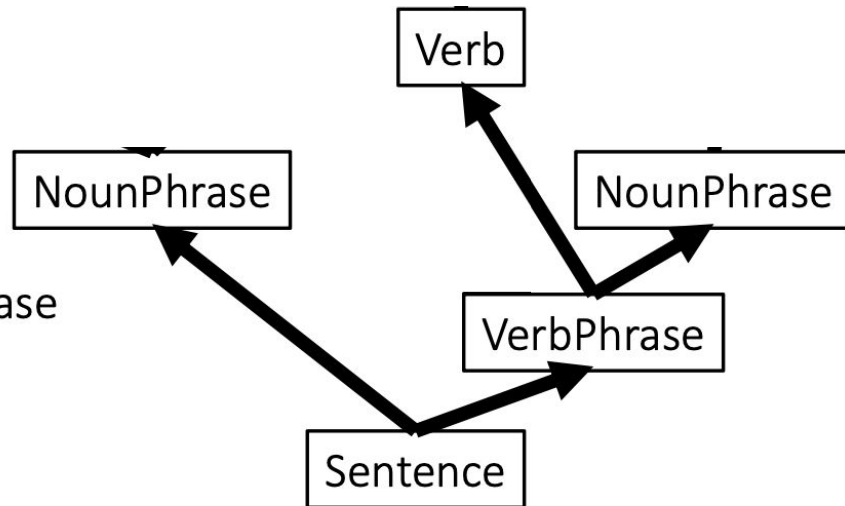
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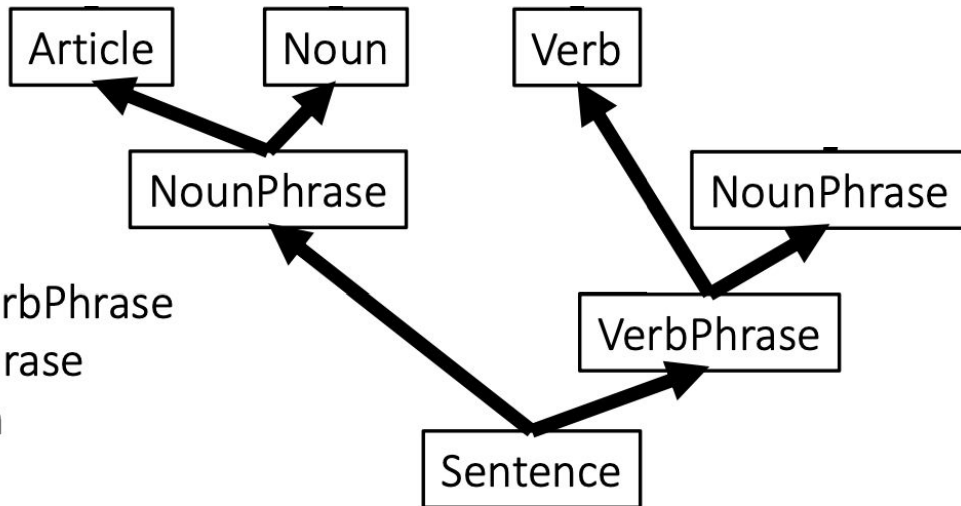
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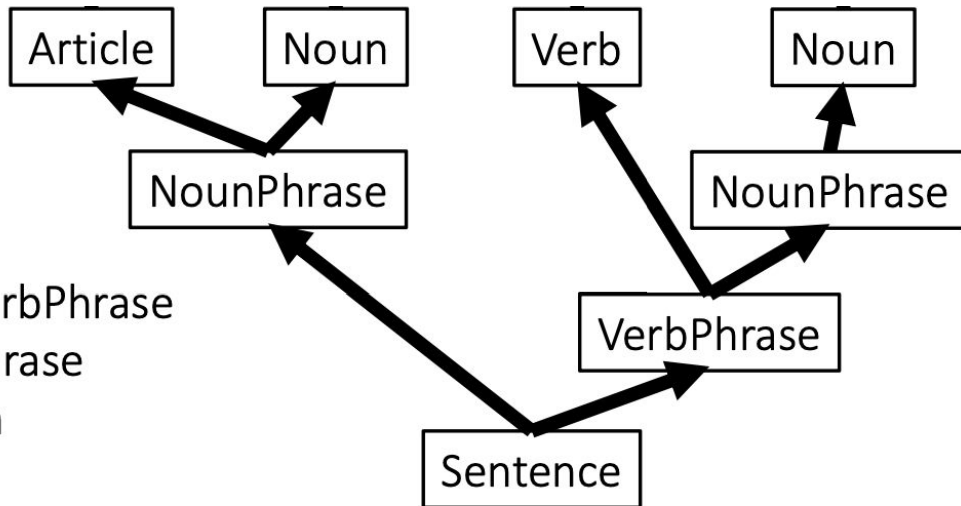
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NounPhrase → Article, Noun

NounPhrase → Noun

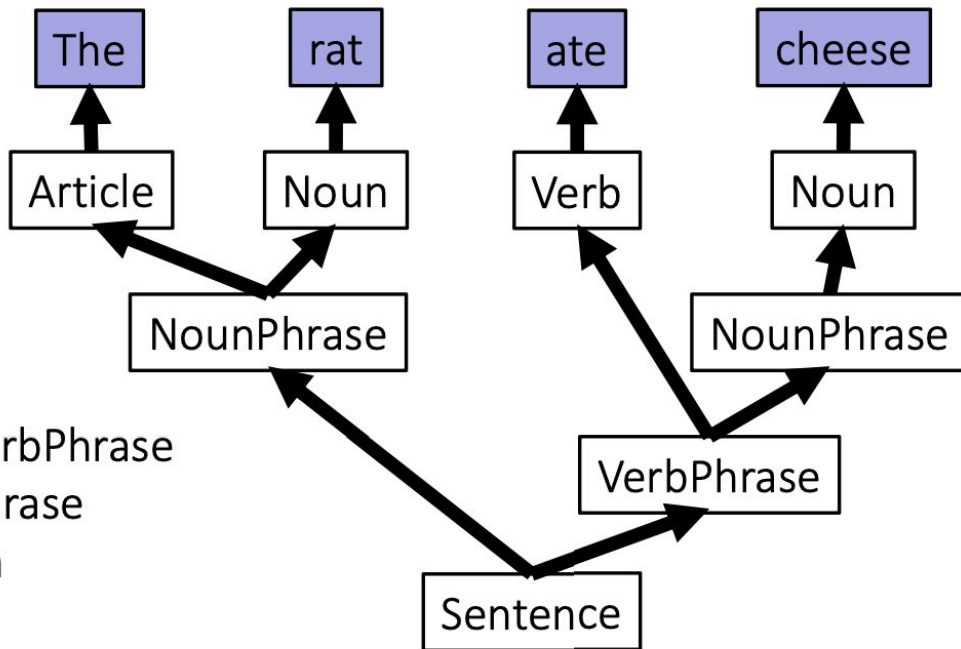


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(4) Output: A parse tree:



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Sentence → NounPhrase, VerbPhrase

VerbPhrase → Verb, NounPhrase

NounPhrase → Article, Noun

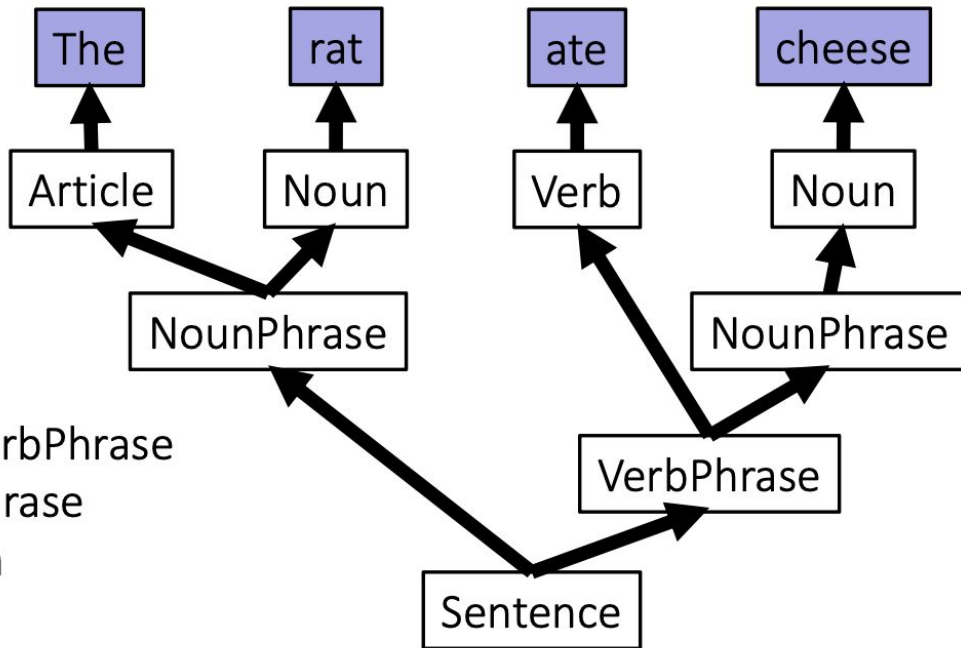
NounPhrase → Noun

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NounPhrase → Article, Noun

NounPhrase → Noun



# Q: Which skill is the most applicable in step 3 (Grammar)?



A. Abstraction

B. Decomposition

C. Synthesis

D. Simulation

E. Modelling

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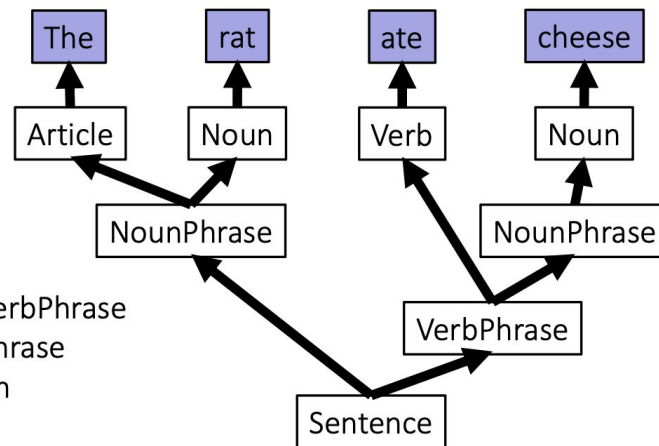
Sentence  $\rightarrow$  NounPhrase, VerbPhrase

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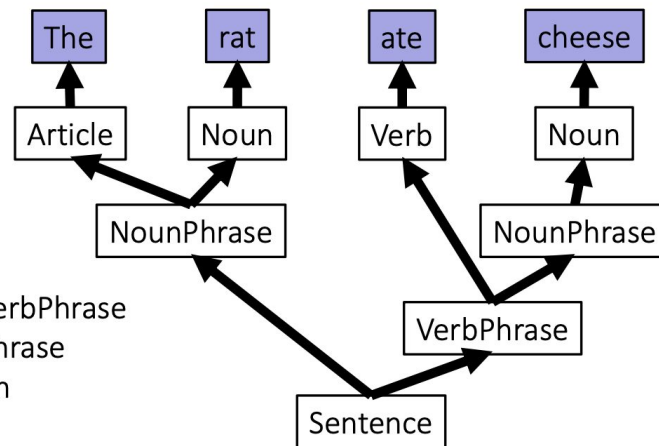
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## Q: Why is the Turing Test significant in AI?

- A. It measures the speed of algorithms.
- B. It assesses whether machines can behave intelligently like humans.
- C. It evaluates the hardware capacity of AI systems.
- D. It rates the efficiency of chatbots.
- E. It assesses whether machines are smarter than humans.



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# Introduction to Natural Language Processing

Natural Language Processing (NLP) is a branch of machine learning focused on enabling computers to understand, interpret, and generate human language.

# Motivation and Context

- Do large language models, such as ChatGPT, “understand” your questions to some extent and provide useful responses?
- What is required for a machine to “understand” language?
- So far we have been talking about sentence or document representations.
- Today, we’ll go one step back and talk about word representations.

# Referential Ambiguity

Let's start with this picture:

If the **baby** does not thrive on raw **milk**, boil it.



**it = ?**



- How do we know what the **it** is referring to?
- How do we tell an algorithm that?



## Ambiguous news headlines

PROSTITUTES APPEAL TO POPE

- **appeal to** means make a serious or urgent request or be attractive or interesting?

KICKING BABY CONSIDERED TO BE HEALTHY

- **kicking** is used as an adjective or a verb?

MILK DRINKERS ARE TURNING TO POWDER

- **turning** means becoming or take up?

# Activity: Context and word meaning

Pair up with the person next to you and try to guess the meanings of two made-up words: **flibbertigibbet** and **groak**.

1. The plot twist was totally unexpected, making it a **flibbertigibbet** experience.
2. Despite its **groak** special effects, the storyline captivated my attention till the end.
3. I found the character development rather **groak**, failing to evoke empathy.
4. The cinematography is **flibbertigibbet**, showcasing breathtaking landscapes.
5. A **groak** narrative that could have been saved with better direction.
6. This movie offers a **flibbertigibbet** blend of humour and action, a must-watch.
7. Sadly, the movie's potential was overshadowed by its **groak** pacing.
8. The soundtrack complemented the film's theme perfectly, adding to its **flibbertigibbet** charm.
9. It's rare to see such a **flibbertigibbet** performance by the lead actor.
10. Despite high expectations, the film turned out to be quite **groak**.
11. **Flibbertigibbet** dialogues and a gripping plot make this movie stand out.
12. The film's **groak** screenplay left much to be desired.

Attribution: Thanks to ChatGPT 4o on Wed Nov. 6, 2024!

## Example: Word similarity

- Suppose you are carrying out sentiment analysis.
- Consider the sentences below.

S1: This movie offers a **flibbertigibbet** blend of humour and action, a must-watch.

S2: This movie offers a **delightful** blend of humour and action, a must-watch.

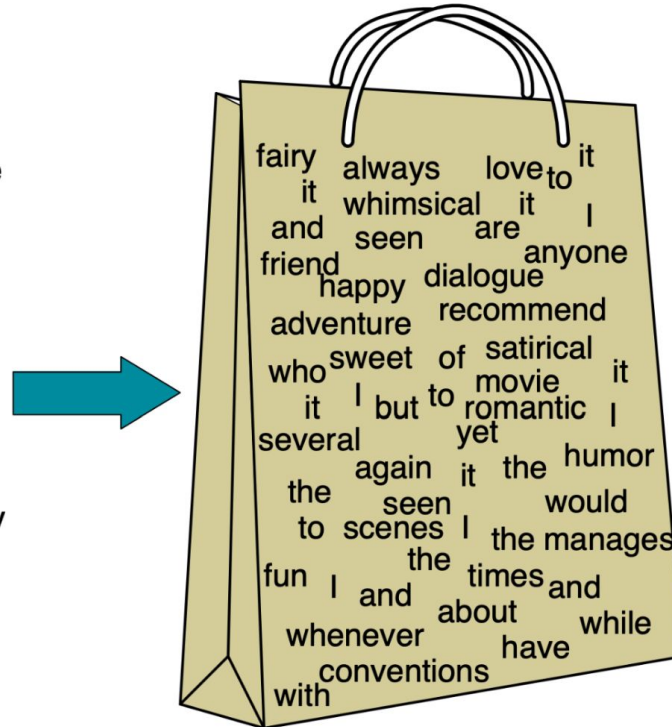
- Here we would like to capture similarity between **flibbertigibbet** and **delightful** in reference to sentiment analysis task.



# Bag of words (BOW) representation

- One of the most popular representation of raw text
- Ignores the syntax and word order
- It has two components:
  - The vocabulary (all unique words in all documents)
  - A value indicating either the presence or absence or the count of each word in the document.

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!

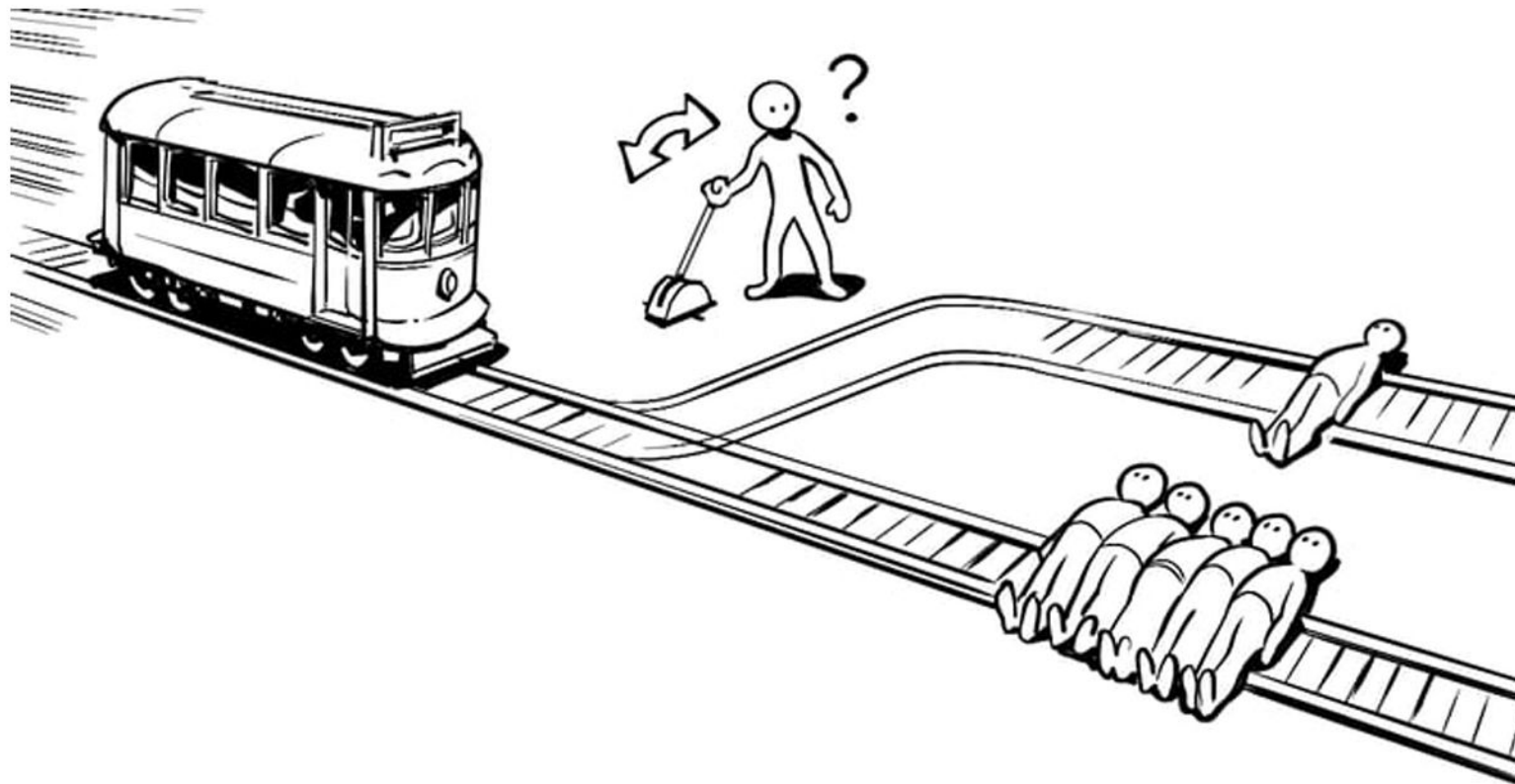


|           |     |
|-----------|-----|
| it        | 6   |
| I         | 5   |
| the       | 4   |
| to        | 3   |
| and       | 3   |
| seen      | 2   |
| yet       | 1   |
| would     | 1   |
| whimsical | 1   |
| times     | 1   |
| sweet     | 1   |
| satirical | 1   |
| adventure | 1   |
| genre     | 1   |
| fairy     | 1   |
| humor     | 1   |
| have      | 1   |
| great     | 1   |
| ...       | ... |

# Trolley Problem



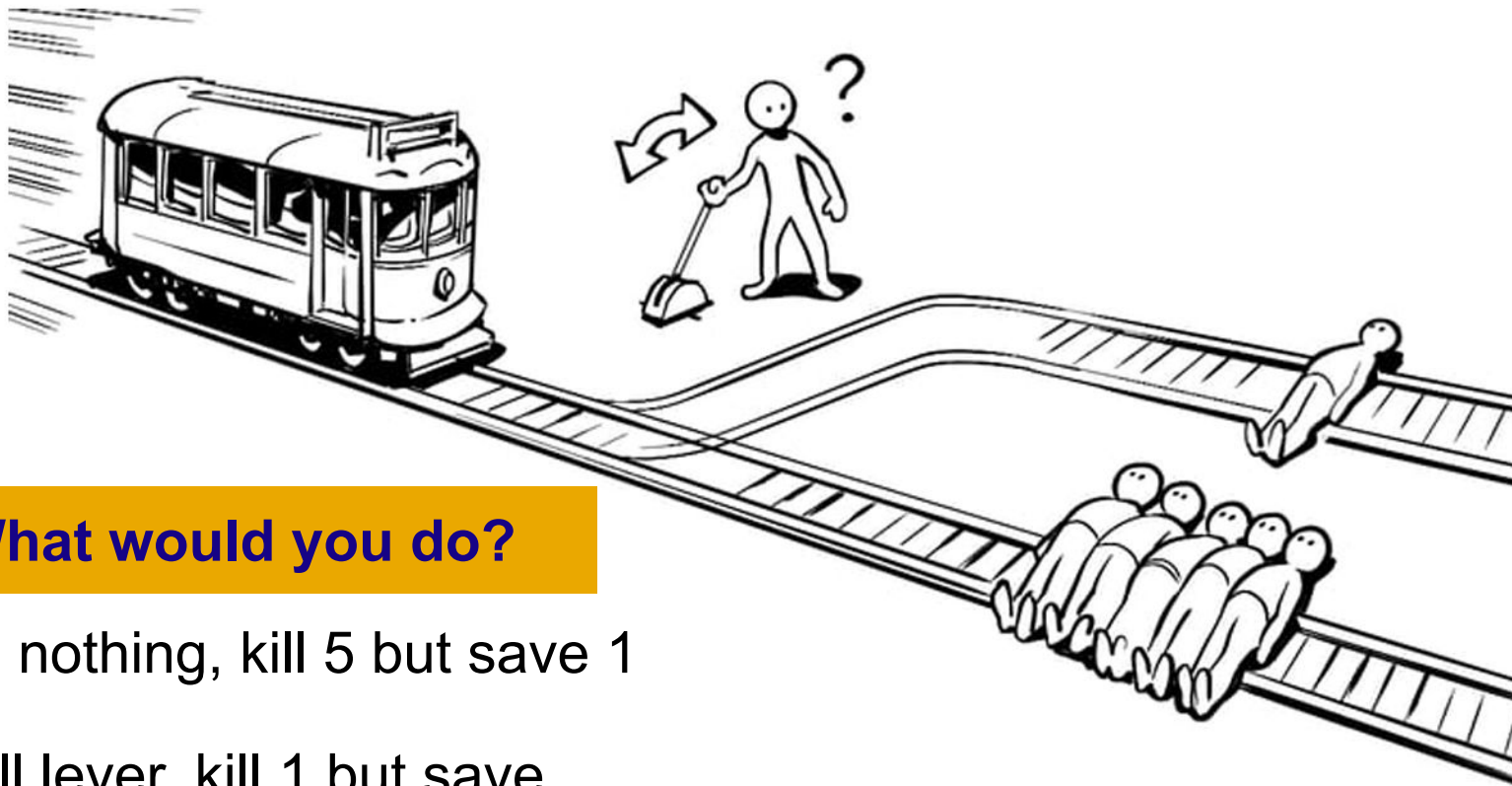
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iClicker



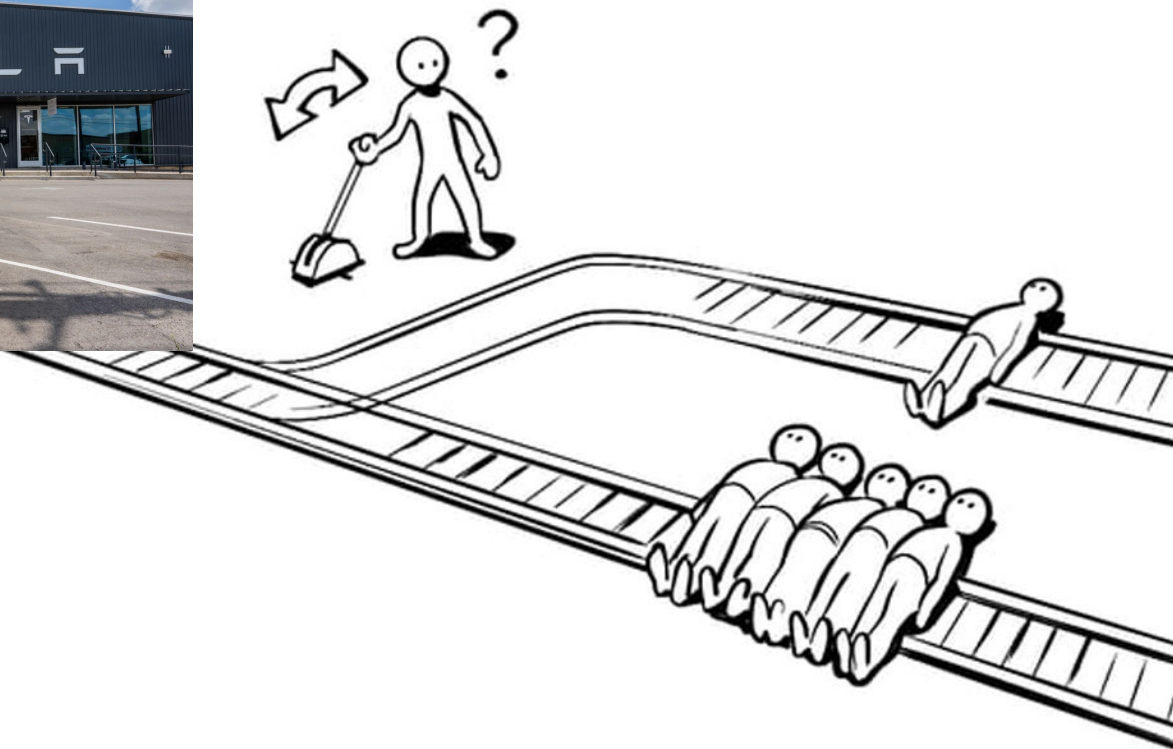
**Q: What would you do?**

- A. Do nothing, kill 5 but save 1
- B. Pull lever, kill 1 but save

ETHICS 101  
 Professor Chirzi Anagonye  
 Trolley Problem - Philippa Foot 1967  
 Deontology vs. Utilitarianism  
 Key Problems:  
 - THE LOOP (COSTA, 1987)  
 - THE MAN IN THE YARD (JENSEN 1992)



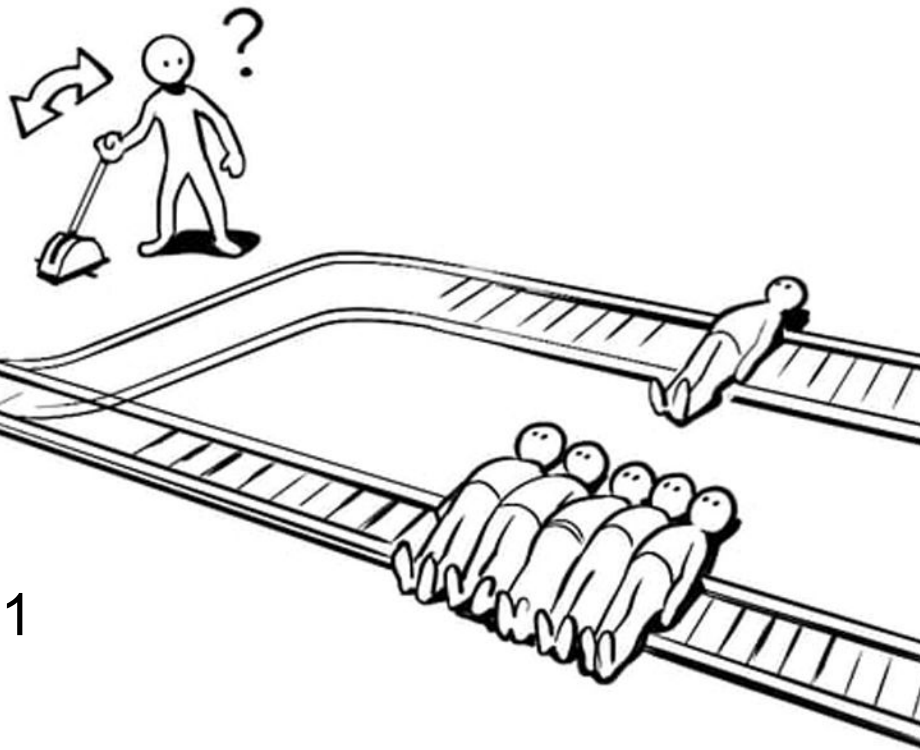
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# Wrap up